Chlorzoxazone PBPK model: fit PK

1 Methods: Chlorzoxazone PBPK model: fit PK

1.1 Parameters and data

7 parameters were optimized: absorption rate chlorzoxazone (`GU__Ka_abs_czx`), dissolution rate czx tablet (`Ka_dis_tablet_czx`), Vmax of the chlorzoxazone 6-hydroxylation (`LI__CZX0X_Vmax`), Km of the chlorzoxazone 6-hydroxylation (`LI__CZX0X_Km`), rate constant of the CYP2E1 independent elimination (`LI__CZX0THER_k`), rate constant of the glucuronidation of 6-hydroxychlorzoxazone (`LI__CZX0GLU_k`) and renal clearance of the glucuronide relative to the GFR (`KI__CZX0GLU_EX_f`).

A fit mapping is a curve of a study, the data and the observable of the model it is compared with. The fit has 54 fit mappings of training data with 414 data points, whose residuals enter the objective; 85 fit mappings of validation data with 727 data points, which were simulated and evaluated with the fitted parameters but not fitted; and 5 fit mappings of outliers with 44 data points, data the fit dropped as not usable, which were evaluated like the validation data so that the decision to drop them can be checked. The fit was set up and run with sbmlsim [1].

1.2 Objective function

The parameters were estimated by minimizing the objective function

$$F(\vec{p}) = \frac{1}{2} \sum_k \sum_i \rho\left(w_{i,k} r_{i,k}(\vec{p})^2\right) \tag{1}$$

where k runs over the fit mappings of the training data and i over their data points, $r_{i,k}$ is the residual, $w_{i,k} = w_k w^p_{i,k}$ the weight of the data point and ρ the loss function, applied to the squared weighted residual $z = w_{i,k} r^2_{i,k}$. The loss function is linear, $\rho(z) = z$, i.e. a standard least squares problem.

The residuals are normalized by the mean $\overline{y}_k$ of the data of their curve, which makes curves of different quantities and magnitudes comparable:

$$r_{i,k} = \frac{y_{i,k} - f_{k(x_{i,k},\vec{p})}}{\overline{y}_k} \tag{2}$$

The weight of curve k is $w_k = \frac{n_k}{N_k}$, the weight n_k of its fit mapping (the weight its fit mapping collection assigns, 1 by default, or, for a collection which uses the weights of its mappings, the weight of the mapping, by default the number of subjects of its data) divided by the number N_k of its data points.

A data point is weighted with the inverse of its coefficient of variation, $w^p_{i,k} = |y_{i,k} \frac{1}{\sigma_{i,k}}$, with $\sigma_{i,k}$ the standard deviation of the data point, or its standard error for data without a standard deviation. An error of zero is missing, a missing error of a curve with errors is replaced by the largest error of the curve, and the data points of a curve without errors have the weight $w^p_{i,k} = 2$, the weight of a coefficient of variation of 0.5.

1.3 Optimization

The optimizer searched the parameters in log10 space within their bounds; bounds, values and intervals are given in the units of the fit parameters, the units of the table of the parameters.

The objective was minimized in 3 runs with differential evolution [2] of SciPy [3] within the bounds of the parameters, with the strategy `best1bin`, a population of 15 times the number of parameters, at most 1000 generations, the relative tolerance 0.01 of convergence, the recombination constant 0.7 and a mutation dithered between 0.5 and 1. The initial population of a run was drawn by latin hypercube sampling [4], with a seed of every run derived from the seed 1234 of the fit. The best member of the final population was polished with the L-BFGS-B method of SciPy.

The convergence of the runs was assessed from the traces of their cost and the waterfall plot of their final costs, and the parameter set with the lowest cost was used.

The models were simulated with libRoadRunner [5] and its integrator CVODE [6] with the relative tolerance 10^{-6} and the absolute tolerances 10^{-12} of an amount, 10^{-12} of a concentration and 10^{-6} of the other states, which for a species in a concentration is multiplied by the initial volume of its compartment. It ran with the variable step size of libRoadRunner, i.e. with an output at every step of the integrator. A data point at the time of a change after the start of a simulation, e.g. a dose or the reset of a collection, was compared with the state before the change, except the first

data point of a curve with data at later times, which starts a collection and was compared with the state after the change.

1.4 Goodness of fit

The goodness of fit is given per kind of data (training data, validation data and outliers) by metrics over its n data points with the residuals $e_i = y_i - f\left(x_i, \hat{\vec{p}}\right)$ at the fitted parameters $\hat{\vec{p}}$, the mean squared error, its root and the root mean squared error normalized by the mean $\overline{y}_{k(i)}$ of the data of the curve of a point:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{NRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{\overline{y}_{k(i)}}\right)^2} \tag{3}$$

The root mean square of the residuals of the objective, i.e. what the fit minimizes, and the coefficient of determination:

$$\mathrm{RMSE}_w = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\rho(w_i r_i^2)}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i \left(y_i - \overline{y}\right)^2} \tag{4}$$

The predictions of a non-linear model are no linear regression of the data, so R^2 can be negative: the prediction is worse than the mean of the data. The information criteria of Akaike [7] and Schwarz [8] with the number n_p of fitted parameters are those of a least squares fit with normally distributed residuals, up to an additive constant, so only differences between models fitted to the same data are meaningful:

$$\mathrm{AIC} = n\ln(\mathrm{MSE}) + 2n_p, \quad \mathrm{BIC} = n\ln(\mathrm{MSE}) + n_p\ln(n) \tag{5}$$

The percentage of the data points whose prediction $f_i = f\left(x_i, \hat{\vec{p}}\right)$ lies within twofold of the data is taken over the n_+ data points with positive data $y_i > 0$ and a positive prediction $f_i > 0$; the others have no ratio and are not counted, and a kind of data without such a point has no percentage:

$$P_2 = \frac{100}{n_+}\left|\left\{i : \frac{1}{2} \le \frac{f_i}{y_i} \le 2\right\}\right| \tag{6}$$

The goodness of fit plot shows the predictions against the data on logarithmic axes, with the lines of twofold $f = 2y$ and $f = \frac{y}{2}$. The Bland-Altman plot [9] shows the ratio of prediction and data as $d_i = \log_{10}\left(\frac{f_i}{y_i}\right)$ against the geometric mean $\sqrt{f_i y_i}$ for the data points with positive values, with the bias $\overline{d}$ and the limits of agreement $\overline{d} \pm 1.96 s_d$, with the sample standard deviation s_d of d_i, of the training data; they are read as the fold factors $10^{\overline{d}}$ and $10^{\overline{d} \pm 1.96 s_d}$.

1.5 Identifiability

The identifiability of the parameters was analysed with the profile likelihood [10], [11]. The profile of a parameter is the lowest objective with the parameter fixed:

$$\mathrm{PL}(p_j) = \min_{p_{i \ne j}} F(\vec{p}) \tag{7}$$

At every point of a scan the other parameters were re-optimized from the previous point with the least squares optimizer of SciPy with the relative step 0.05. A scan ran from the optimum in both directions in the space of the optimizer, with steps in decades of the parameter, starting at 0.1. A step which raised the objective by more than 0.2 of the distance between the objective at the optimum and the threshold was reduced by the factor 2 and repeated, down to 0.01, and the step after one which raised it by less than that divided by the factor was enlarged by the factor, up to 1. A scan stopped at the threshold, at the bound of the parameter, at a point which failed or after 50 points.

The confidence intervals are the values at which the profile crosses the threshold of the likelihood ratio test

$$F_{\mathrm{thr}} = F_{\mathrm{min}} + \chi^2_\alpha \frac{\mathrm{df}}{2} \tag{8}$$

with F_{min} the lowest objective of the fit and of all profiles and $\chi^2_\alpha(\mathrm{df})$ the quantile of the χ^2 distribution at the confidence level $\alpha = 0.95$ with df = 1 degrees of freedom, i.e. 1.92 above the lowest objective. The threshold has this confidence level if F is the negative log-likelihood of the data up to a constant, i.e. if the weighted residuals are independent and normally distributed with unit variance; otherwise the level is approximate. A crossing is interpolated between two points of the profile, on the logarithm of a parameter on a logarithmic scale. A parameter is identifiable if its profile reaches the threshold within the scan on both sides of the optimum, i.e. its confidence

interval is finite; practically non-identifiable towards the lower bound if its profile reaches the threshold above the optimum but not below it within the scan, which ends at the lower bound or after 50 points; practically non-identifiable towards the upper bound if its profile reaches the threshold below the optimum but not above it within the scan, which ends at the upper bound or after 50 points; practically non-identifiable if its profile reaches the threshold on neither side within the scans, which end at the bounds or after 50 points, and is not flat; and structurally non-identifiable if its profile is flat on both sides. A side of a profile is flat if it rises by less than 0.05 of the distance between the lowest objective and the threshold, and a profile with a point which failed or did not converge is not classified. The profile likelihood is preferred over the intervals of the Fisher information, its intervals are invariant under a transformation of the parameters and can be asymmetric [12].

The Fisher information is the local alternative to the profiles: it is the Gauss-Newton approximation of the curvature of the objective at the optimum, from the Jacobian J of the weighted residuals of the training data with respect to the parameters $\vec{p}$ in the space of the optimizer, computed by central finite differences with the step $h_j = \varepsilon \max\left(|p_j|, 1\right),\ \varepsilon = 10^{-6}$ of the parameter p_j:

$$I = J^{\top} J, \quad \mathrm{cov} = \sigma^2 I^{-1}, \quad \sigma^2 = \frac{2F}{n - n_p} \tag{9}$$

with the number n of data points of the training data and the number n_p of parameters. The information has rank 7 of 7, the number of its eigenvalues above 10^{-8} of the largest one, and the condition number 92.3; the covariance of an information of a lower rank is computed with the pseudo-inverse. The Wald intervals are $\hat{p}_j \pm t_{\frac{1+\alpha}{2}, n-n_p} \mathrm{SE}_j$ with the standard errors $\mathrm{SE}_j = \sqrt{\mathrm{cov}_{jj}}$ at the confidence level $\alpha = 0.95$, transformed back into the units of the fit parameters.

1.6 Provenance

The fit PK_20261020_035132__b4a12 was run on 2026-10-20 01:51:35 UTC with sbmlsim 0.10.0 [1], libRoadRunner 2.10.0 [5], SciPy 1.18.1 [3], NumPy 2.5.3 [13] and Python 3.14.4 (seed 1234). The models and data were those of commit `9223ef2853a750b781a98a244c84bf997b40fdec` of `/home/mkoenig/git/pkdb_models-chlorzoxazone` their SHA-256 checksums are listed in Table 1.

Table 1: SHA-256 checksums of the models and data of the fit PK_20261020_035132__b4a12, their first 12 characters; the full checksums are in the provenance of the result (optimization_result.json).

File	SHA-256
`pkdb_models/models/chlorzoxazone/data/alcohol/Wilkinson1977/.Wilkinson1977_Fig1_2.tsv`	`8ddacf987211...`
`pkdb_models/models/chlorzoxazone/data/alcohol/Wilkinson1977/.Wilkinson1977_TabGroups.tsv`	`f0bd8ceea2f7...`
`pkdb_models/models/chlorzoxazone/data/alcohol/Wilkinson1977/.Wilkinson1977_TabIndividuals.tsv`	`612f98e8c8be...`
`pkdb_models/models/chlorzoxazone/data/alcohol/Wilkinson1977/.Wilkinson1977_TabInterventions.tsv`	`ec349ebdb2f4...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Fig1.tsv`	`35719eecf965...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Fig2.tsv`	`5b0d2edccef9...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Fig3.tsv`	`f330cc9a69d8...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Fig4.tsv`	`40f066a32390...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Tab1.tsv`	`5d79b99fb2d8...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_Tab2.tsv`	`d7d3f42b70f1...`
`pkdb_models/models/chlorzoxazone/data/caffeine/Benowitz2003/.Benowitz2003_TabData.tsv`	`75037e3b2696...`
`pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_Fig1.tsv`	`b12fd2723098...`

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_Fig2.tsv	acf5598ef022...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_Tab1.tsv	34c1cc729100...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_Tab2.tsv	f8179d01b2b9...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_TabGroups.tsv	e7c507e0d746...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2016/.Bedada2016_TabInterventions.tsv	6bf441696386...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_Fig2.tsv	0f4d53c2c6c5...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_Fig3.tsv	0a74c082763e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_Fig4.tsv	d7853b088236...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_Tab1.tsv	f2976ed54bc8...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_Tab2.tsv	34c961912e0b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_TabGroups.tsv	5f0dc23e1e32...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2017/.Bedada2017_TabInterventions.tsv	b0508f97cb84...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2018/.Bedada2018_Fig1.tsv	2c8b9af4bd98...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2018/.Bedada2018_Fig2.tsv	15a98a490926...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2018/.Bedada2018_Tab1.tsv	9b510fe4a0b3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2018/.Bedada2018_TabGroups.tsv	4b41e580e269...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Bedada2018/.Bedada2018_TabInterventions.tsv	d7e76ded8584...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_Fig1.tsv	8f874ea97cbb...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_Fig2.tsv	8f31af2316c2...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_Fig4.tsv	baf769dd7739...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_Tab2.tsv	124a16328593...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_TabGroups.tsv	b21a3b85bb25...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Burckart1998/.Burckart1998_TabInterventions.tsv	f48b88b96fa3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Chalasani2003/.Chalasani2003_Fig1.tsv	90eb6e644305...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Chalasani2003/.Chalasani2003_Tab1.tsv	eb854d69f070...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Chalasani2003/.Chalasani2003_TabGroups.tsv	f687e66726c7...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Chalasani2003/.Chalasani2003_TabInterventions.tsv	071faf1c14fd...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_Fig1.tsv	653c6463393e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_Fig2.tsv	a18d18bbc128...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_Tab2.tsv	910906e27162...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_TabGroups.tsv	35497c2b34cf...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_TabIndividuals.tsv	93783d735f6b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/DeVries1994/.DeVries1994_TabInterventions.tsv	b69aae9e3476...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_Fig1.tsv	eb9ae4cd8916...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_Fig2.tsv	9eb245d08235...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_Fig3.tsv	74e8510bfcfd...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_Tab1.tsv	76494e4df99d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_TabGroups.tsv	d5da7c3849ca...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Desiraju1983/.Desiraju1983_TabInterventions.tsv	42d17442e8a8...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Fig1.tsv	9c906834a466...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Fig2.tsv	a920aae60fac...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Fig3.tsv	17b17e4316fc...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Fig4.tsv	d00ab57bf7d4...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Tab1.tsv	3ad8cde7c399...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_Tab4.tsv	16b6824fff64...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_TabGroups.tsv	c41fd19baa0b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Dreisbach1995/.Dreisbach1995_TabInterventions.tsv	aae1f33221d0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Eap1998/.Eap1998_Fig1.tsv	786e3fffbed0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Eap1998/.Eap1998_Tab1.tsv	689035d63548...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Eap1998/.Eap1998_TabGroups.tsv	d1f4d29d6b4c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Eap1998/.Eap1998_TabInterventions.tsv	08852bcff644...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Emery2003/.Emery2003_Fig1.tsv	ddf8707ee298...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Emery2003/.Emery2003_Fig4.tsv	c0e1a1717025...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Emery2003/.Emery2003_Tab1.tsv	627aa91bbffa...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Emery2003/.Emery2003_TabGroups.tsv	028bd69ce9cc...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Emery2003/.Emery2003_TabInterventions.tsv	f26264a66b84...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_Fig1.tsv	8a8aeaed3b47...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_Fig5.tsv	7e1e68b815a0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_Tab2.tsv	2866e7e67134...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_TabGroups.tsv	92db4f6deb27...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_TabIndividuals.tsv	c5a11b9a98b7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Ernstgard2004/.Ernstgard2004_TabInterventions.tsv	dd5474690c6e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig1A.tsv	63e322211228...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig1B.tsv	380b2c5eb263...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig2A.tsv	94a317187a10...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig2B.tsv	89a0e9ffdfa3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig3A.tsv	aebc858b82c5...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig3B.tsv	7aea22b42c92...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Fig4.tsv	e9c8158286a3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Tab1.tsv	cc0fb1f523a7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_Tab3.tsv	00a745b66226...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_TabGroups.tsv	d4a08c4c6bc8...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Frye1998/.Frye1998_TabInterventions.tsv	64d59d6ce61d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Gade2018/.Gade2018_Fig1.tsv	b81e4f66a8bc...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Gade2018/.Gade2018_Tab2.tsv	eb2ed1721011...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Gade2018/.Gade2018_TabGroups.tsv	5d931df6332c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Gade2018/.Gade2018_TabInterventions.tsv	c375d9a29f43...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_Fig1.tsv	d15a7cb0b2fb...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_Fig2.tsv	f56ff51daff1...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_Fig3.tsv	2bd68381876f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_Tab2.tsv	15f96664a259...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_TabGroups.tsv	3a65e4ec9bf9...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Girre1994/.Girre1994_TabInterventions.tsv	efa5d5d9ed80...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/He2019/.He2019_Fig4.tsv	d4535c11a47c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/He2019/.He2019_Tab4.tsv	7620b413d5ea...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/He2019/.He2019_TabGroups.tsv	9d4b32a3293a...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/He2019/.He2019_TabInterventions.tsv	81c9048a4a91...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hohmann2019/.Hohmann2019_Fig2A.tsv	9bd45d3d0e87...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hohmann2019/.Hohmann2019_Fig2B.tsv	1e39abe5e6d3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hohmann2019/.Hohmann2019_Tab1.tsv	21652a66eba8...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hohmann2019/.Hohmann2019_TabGroups.tsv	1db0190675d2...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hohmann2019/.Hohmann2019_TabInterventions.tsv	2aeea2424a5b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hukkanen2010/.Hukkanen2010_Tab4.tsv	dc56ad9b4cd4...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hukkanen2010/.Hukkanen2010_Tab5.tsv	307d3a53bfbf...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hukkanen2010/.Hukkanen2010_TabGroups.tsv	a31685e1cdca...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Hukkanen2010/.Hukkanen2010_TabInterventions.tsv	7af91fd4fa16...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_Fig2.tsv	92721940508e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_Tab1.tsv	d1679015ce86...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_Tab1Mean.tsv	eb814c79ed7b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_Tab2.tsv	330e50823c12...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_Tab2Mean.tsv	382b7bcf168e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_TabGroups.tsv	9088d72c9cd5...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_TabIndividuals.tsv	f4a7724bd56a...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Kharasch1993/.Kharasch1993_TabInterventions.tsv	ce9482996002...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_Fig1.tsv	512ce5f3107f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_Fig2.tsv	f3cf5a269a4c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_Fig3.tsv	f55a53041f9a...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_Tab3.tsv	99064c8e3eb2...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_TabGroups.tsv	58e4cefe4968...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/LaMaza2000/.LaMaza2000_TabInterventions.tsv	34af6a0226a3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Leclercq1998/.Leclercq1998_Fig1.tsv	d9adc115aa7f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Leclercq1998/.Leclercq1998_Tab1.tsv	039964865778...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Leclercq1998/.Leclercq1998_Tab2.tsv	6749855cb6de...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Leclercq1998/.Leclercq1998_TabGroups.tsv	fb150d12bdc8...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Leclercq1998/.Leclercq1998_TabInterventions.tsv	a5b008d948ed...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_Fig1.tsv	d088fe578cd0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_Fig2.tsv	31d8eeb0e6e7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_Tab2.tsv	12139755ed9b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_Tab3.tsv	e40129208571...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_TabGroups.tsv	a95081b27503...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Liangpunsakul2005/.Liangpunsakul2005_TabInterventions.tsv	35c66c3df6d6...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1993/.Lucas1993_Fig2.tsv	b04dc3282f5e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1993/.Lucas1993_Tab5.tsv	46e1843438b3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1993/.Lucas1993_TabGroups.tsv	358aa6e9e71a...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1993/.Lucas1993_TabInterventions.tsv	4cb378baa0a5...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/.Lucas1995_Fig1.tsv	c3b3d9280881...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/.Lucas1995_Tab2.tsv	5d81ad044dff...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/.Lucas1995_TabGroups.tsv	aca43fc3f875...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/.Lucas1995_TabIndividuals.tsv	758cf65c03cc...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/.Lucas1995_TabInterventions.tsv	d3279eebb93d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Lucas1995/Lucas1995.xlsx	b04daba84907...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig1.tsv	7b2565f9b2b9...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig2.tsv	29ed828c669f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig3.tsv	5a6a8a771c1a...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig4.tsv	58c6b0c08f1b...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig5.tsv	31b43e10a5d7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig6.tsv	f320ae266b8e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig7.tsv	00837fb0aafb...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig8.tsv	0d13bea44f8c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Fig9.tsv	505d7003133d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_Tab2.tsv	79037b4b7b75...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_TabGroups.tsv	0be20fc1e664...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Mishin1998/.Mishin1998_TabInterventions.tsv	7bb8877a7c8b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Fig1.tsv	cfd87278e6f3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Fig2.tsv	178f920196a7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Tab1_2.tsv	71c55082fa93...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Tab1_2Mean.tsv	940d84f74d00...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Tab3.tsv	2d3e369422b1...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_Tab3Mean.tsv	943d4ac21d8f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_TabGroups.tsv	4e961328aa65...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_TabIndividuals.tsv	c8fe479027eb...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/OShea1994/.OShea1994_TabInterventions.tsv	7acd3a99df5f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/.Oneta2002_Fig2.tsv	428d17c46c62...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/.Oneta2002_Fig3.tsv	6d36b8ff0481...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/.Oneta2002_Fig3_NALD.tsv	a5392307b7a0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/.Oneta2002_TabGroups.tsv	dad65d70efba...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/.Oneta2002_TabInterventions.tsv	3232f7c689b4...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Oneta2002/Oneta2002.xlsx	998b34d57ac3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Orellana2006/Orellana2006.xlsx	b048457c6631...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Park2006/.Park2006_Fig1.tsv	063a38f0b548...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Park2006/.Park2006_Tab1.tsv	e46a57a5bea2...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Park2006/.Park2006_TabGroups.tsv	953475d9b509...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Park2006/.Park2006_TabInterventions.tsv	948d0cd3b13f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_Fig1.tsv	ac419cf0e1c0...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_Tab2.tsv	118f0088ed72...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_TabGroups.tsv	b518933db2c3...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_TabInterventions.tsv	1574a274f2fa...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_TabS3.tsv	e08de745db7c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Puris2019/.Puris2019_TabS4.tsv	61f03a6b5f6d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Rajnarayana2008/.Rajnarayana2008_Fig1.tsv	c1127df5db53...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Rajnarayana2008/.Rajnarayana2008_Tab1.tsv	017d2ed04d1d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Rajnarayana2008/.Rajnarayana2008_Tab3.tsv	90d004da3fc9...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Rajnarayana2008/.Rajnarayana2008_TabGroups.tsv	54d04bad3f7c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Rajnarayana2008/.Rajnarayana2008_TabInterventions.tsv	747aa0e10d5f...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_Fig1.tsv	5f84b1508dc9...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_Fig2.tsv	88fac2b962bc...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_Fig3.tsv	6aee1fd9db21...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_Tab1.tsv	e61fd6608e4e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_TabGroups.tsv	37ddcbf5134d...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_TabIndividuals.tsv	242948e02d7c...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Vesell1995/.Vesell1995_TabInterventions.tsv	d5d4693feb0e...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Wang2003/.Wang2003_Fig1.tsv	a32f82c4f959...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Wang2003/.Wang2003_Fig2.tsv	8f31af2316c2...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Wang2003/.Wang2003_Tab3.tsv	89e079dc9733...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Wang2003/.Wang2003_TabGroups.tsv	8affed456344...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Wang2003/.Wang2003_TabInterventions.tsv	5bb6cedee7b9...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Witt2016/.Witt2016_Fig3.tsv	d0f79e8f6bf7...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Witt2016/.Witt2016_Tab2.tsv	07c32c0d686b...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Witt2016/.Witt2016_TabGroups.tsv	eacd1133a977...
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Witt2016/.Witt2016_TabIndividuals.tsv	686796bf080c...

File	SHA-256
pkdb_models/models/chlorzoxazone/data/chlorzoxazone/Witt2016/.Witt2016_TabInterventions.tsv	b86dee72309a...
pkdb_models/models/chlorzoxazone/models/output/chlorzoxazone_body_flat.xml	33557e52012b...

2 Results: Chlorzoxazone PBPK model: fit PK

2.1 Parameters

Table 2: Parameters of the fit PK_20261020_035132__b4a12, the parameter set PK_20261020_035132__b4a12_0, cost 131, with their bounds in the optimization. The 95% confidence intervals are those of the profile likelihood, LB and UB an interval which is open towards the lower or the upper bound, and the Wald intervals of the Fisher information for a parameter without a profile. Identifiable is the classification of the profile: yes, no (LB) or no (UB) if the profile does not reach the threshold towards the lower or the upper bound, no if it reaches it on neither side, structural if it is flat.

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
GU__Ka_abs_czx	absorption rate chlorzoxazone	1.17 [0.985, 1.41]	1/hr	[0.3, 10]	yes
Ka_dis_tablet_czx	dissolution rate czx tablet	0.565 [0.486, 0.662]	1/hr	[0.2, 20]	yes
LI__CZXOX_Vmax	Vmax of the chlorzoxazone 6-hydroxylation	0.00717 [0.00554, 0.00987]	mmol/min/l	[0.0001, 1]	yes
LI__CZXOX_Km	Km of the chlorzoxazone 6-hydroxylation	0.0099 [0.0067, 0.0156]	mmol/l	[0.0001, 0.5]	yes
LI__CZXOTHER_k	rate constant of the CYP2E1 independent elimination	0.263 [0.228, 0.302]	1/min	[0.001, 2]	yes
LI__CZXOGLU_k	rate constant of the glucuronidation of 6-hydroxychlorzoxazone	42.8 [35.9, 51.2]	1/min	[1, 1e+04]	yes
KI__CZXOGLU_EX_f	renal clearance of the glucuronide relative to the GFR	14.6 [11.4, 20]	dimensionless	[0.5, 50]	yes

2.2 Data

Table 3: Data of the fit PK_20261020_035132__b4a12: the number of fit mappings and, in parentheses, of data points per study and kind. The training data enters the cost, the validation data and the outliers are evaluated only.

Study	Training	Validation	Outlier	Total
Bedada2016	1 (10)	-	1 (10)	2 (20)
Bedada2017	1 (10)	-	1 (10)	2 (20)
Bedada2018	1 (10)	-	1 (10)	2 (20)
Benowitz2003	2 (12)	-	-	2 (12)
Burckart1998	3 (9)	-	-	3 (9)
Chalasani2003	-	2 (10)	-	2 (10)
Desiraju1983	2 (15)	-	-	2 (15)
DeVries1994	-	17 (179)	-	17 (179)
Dreisbach1995	1 (5)	2 (22)	-	3 (27)
Eap1998	2 (14)	2 (14)	-	4 (28)

Study	Training	Validation	Outlier	Total
Frye1998	6 (69)	-	-	6 (69)
Gade2018	2 (24)	2 (24)	-	4 (48)
Girre1994	3 (27)	-	-	3 (27)
He2019	1 (13)	-	-	1 (13)
Hohmann2019	14 (107)	-	-	14 (107)
Hukkanen2010	1 (1)	-	-	1 (1)
Kharasch1993	1 (1)	3 (20)	-	4 (21)
LaMaza2000	1 (11)	3 (33)	-	4 (44)
Leclercq1998	-	6 (54)	-	6 (54)
Liangpunsakul2005	2 (10)	2 (10)	-	4 (20)
Lucas1993	3 (27)	3 (27)	-	6 (54)
Mishin1998	-	6 (60)	-	6 (60)
OShea1994	1 (1)	8 (72)	-	9 (73)
Park2006	2 (18)	2 (18)	-	4 (36)
Puris2019	1 (4)	1 (4)	-	2 (8)
Rajnarayana2008	-	1 (9)	1 (9)	2 (18)
Vesell1995	1 (5)	6 (40)	1 (5)	8 (50)
Wang2003	2 (11)	4 (22)	-	6 (33)
Witt2016	-	15 (109)	-	15 (109)
Total	54 (414)	85 (727)	5 (44)	144 (1185)

2.3 Goodness of fit

Table 4: Goodness of fit of the fit PK_20261020_035132__b4a12 per kind of data, the parameter set PK_20261020_035132__b4a12_0. The cost is defined on the training data only, 2-fold is the percentage of the data points with a positive value and prediction whose prediction is within twofold of the data.

Kind	n	cost	MSE	RMSE	NRMSE	RMSE_w	R^2	2-fold [%]	AIC	BIC
training	414	131	0.0161	0.127	0.446	0.795	0.933	81.8	−1696.5	−1668.3
validation	727	-	0.00209	0.0457	0.754	0.685	0.948	60.1	−4473.6	−4441.5
outlier	44	-	0.00255	0.0505	0.887	1.69	0.946	27.3	−248.7	−236.2

2.4 Figures

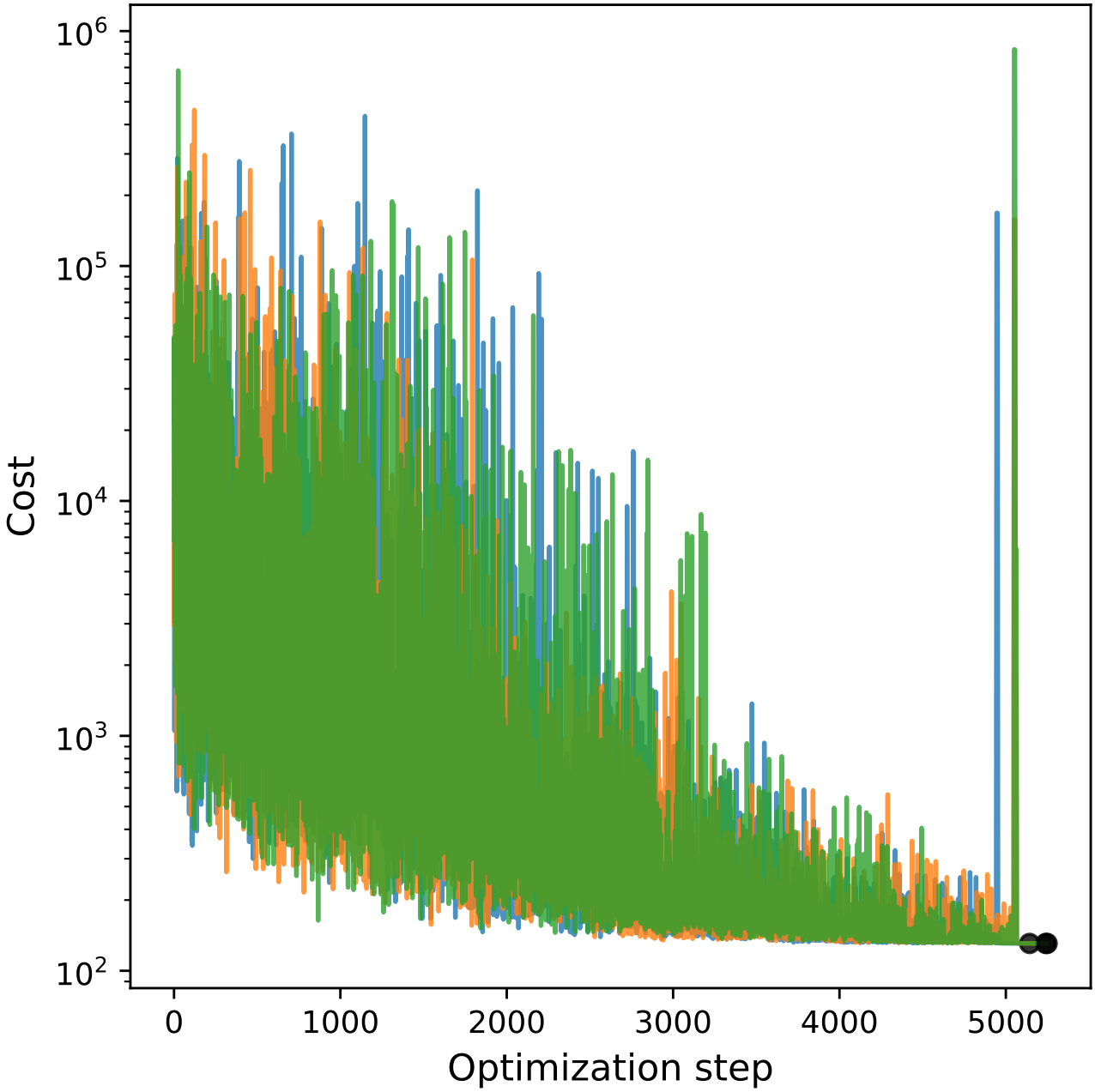


Figure 1: Cost of the optimizers over their steps, fit PK_20261020_035132__b4a12. The cost of every optimization run over its steps. Runs which end at the same cost found the same optimum; a run which stops much higher is stuck in a local minimum.

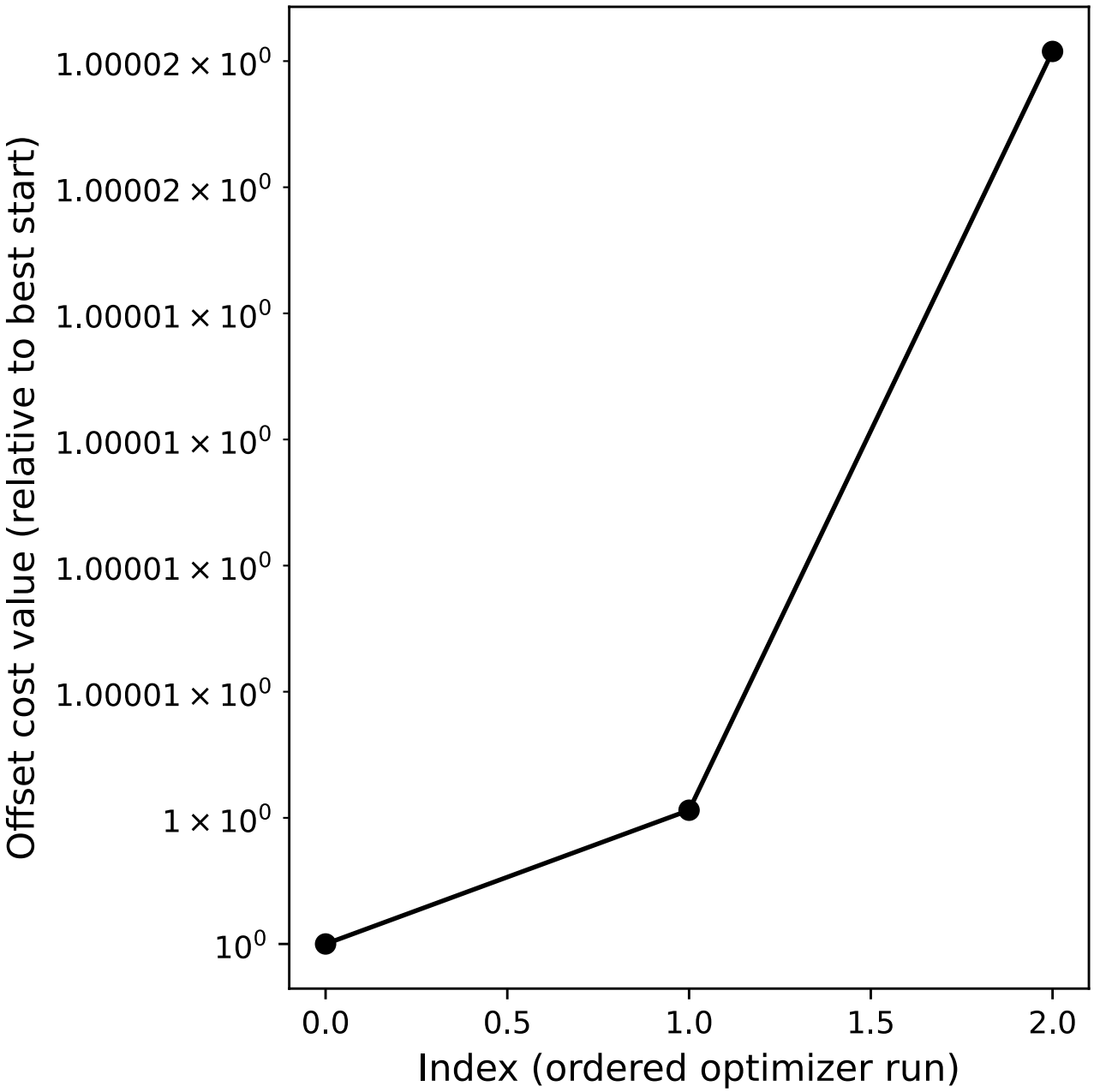


Figure 2: Cost of the optimization runs, ordered, fit PK_20261020_035132__b4a12. The final cost of the runs, ordered. A flat plateau at the left is the global optimum found repeatedly, which is the evidence that the multistart converged; steps to the right are local minima.

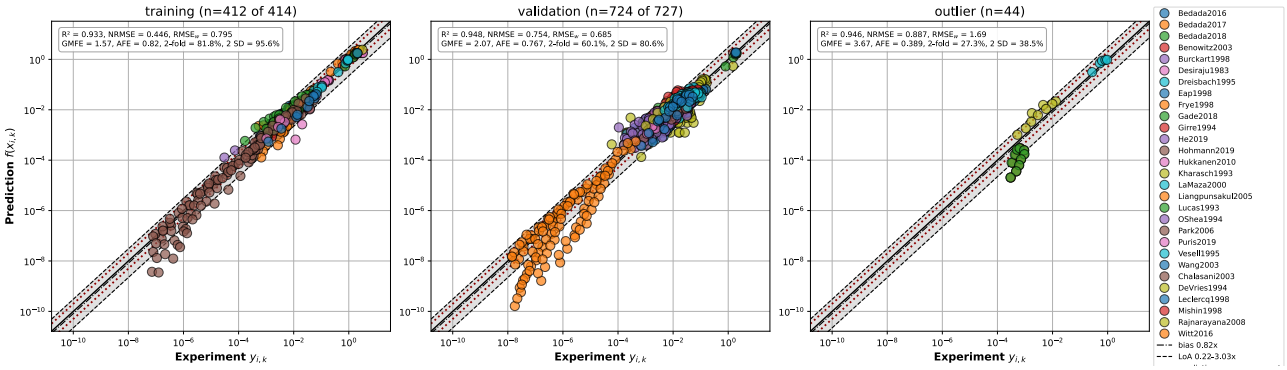


Figure 3: Prediction against the measured data points, per kind of fit mapping, with the agreement of the training data, the lines of twofold and the metrics of the panel, fit PK_20261020_035132__b4a12. Within twofold of the data are 337 of the 412 training data points (81.8%), 435 of the 724 validation data points (60.1%) and 12 of the 44 outlier data points (27.3%); only the data points with a positive value and prediction are counted. Prediction against measurement, one point per data point, with a panel per kind of fit mapping. The points scatter around the diagonal when the model describes the data; a systematic deviation from it is a systematic error of the model. The band is the agreement of the training data, i.e. the same band the Bland-Altman plot draws, here as lines parallel to the diagonal. The dark red dotted lines are twofold, prediction = 2 x measurement and prediction = measurement / 2; the caption states the points between them of every kind. A point which is zero or negative is not drawn on the logarithmic axes and not counted, the title of a panel then states the points drawn of all. The validation panel shows how the fit describes data it was not fitted on, the outlier panel the data the fit dropped. The box of a panel carries its R^2, its normalized RMSE, the RMSE of the residuals of the cost, its fold errors GMFE and AFE and the percentages of its points within twofold and within two standard deviations of the data.

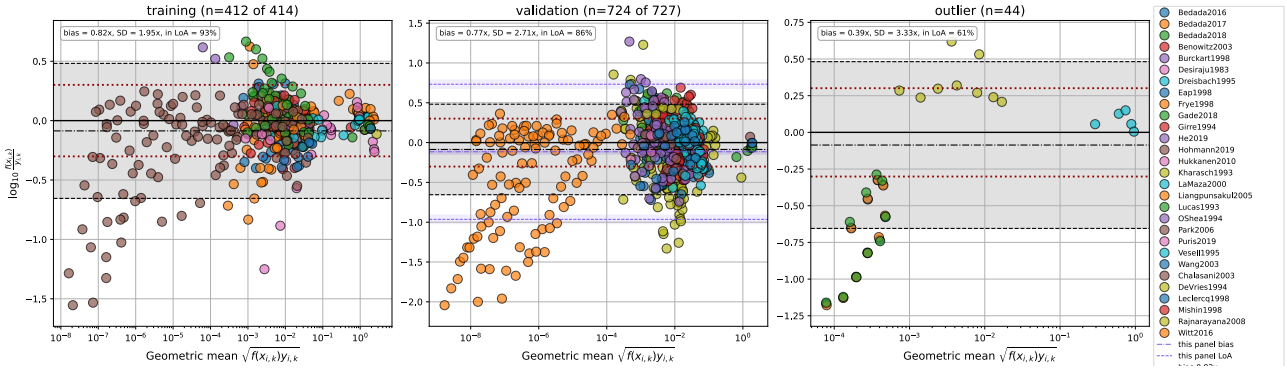


Figure 4: Agreement of prediction and measurement as a ratio, per kind of fit mapping, with the agreement of the training data and the agreement of the panel, fit PK_20261020_035132__b4a12. The ratio of prediction and measurement over the geometric mean of the two, i.e. the goodness of fit with the diagonal turned into the horizontal. The band is the same, the bias and the limits of agreement `bias ± 1.96 SD` of the training data as fold factors. A bias away from 1 is a systematic over- or underprediction, wide limits are a large scatter, and a trend over the mean is a model which describes the large or the small values better. The box of a panel carries the bias and the SD of its own points as fold factors and the share of them inside the limits of agreement of the training data.

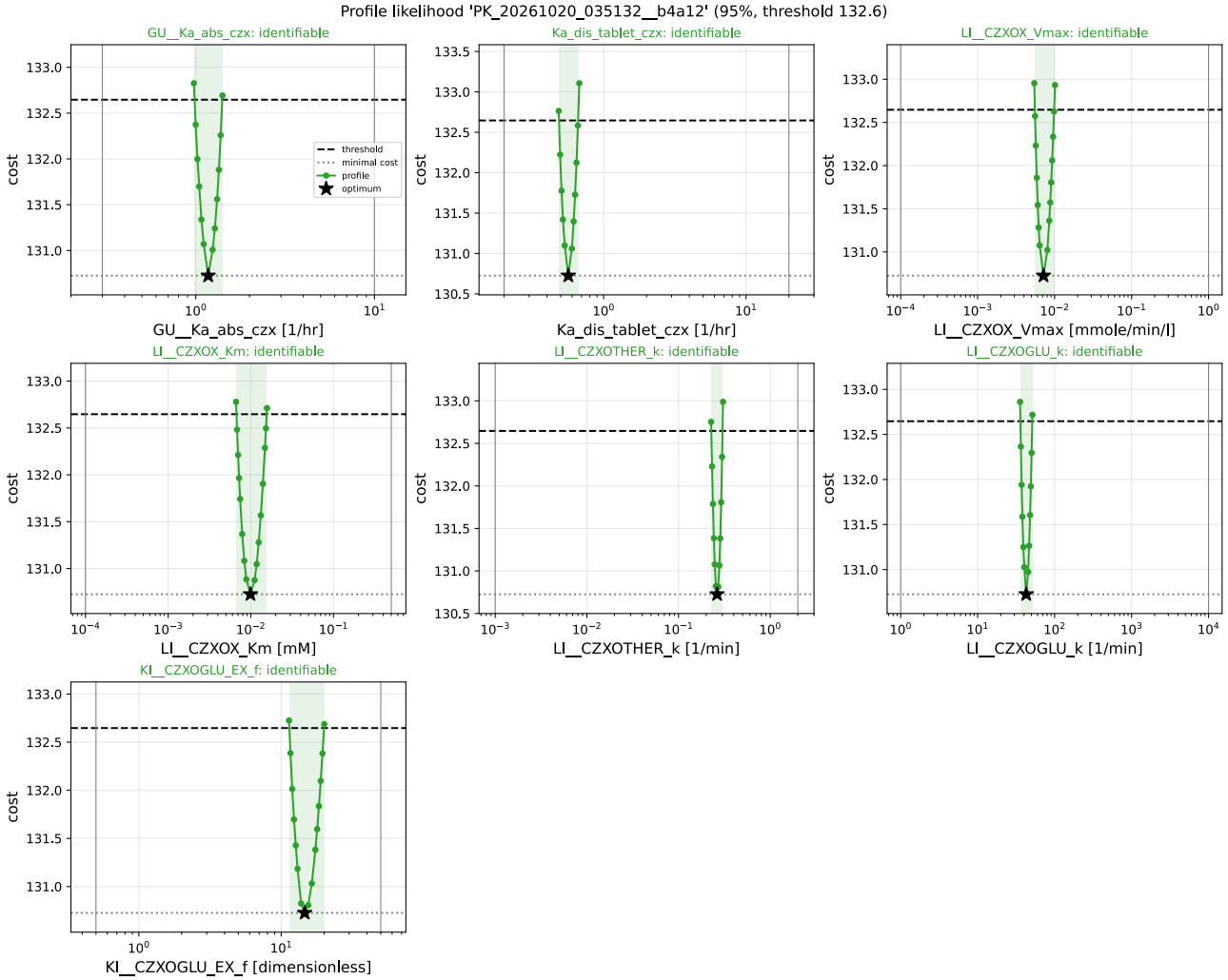


Figure 5: Profile likelihood of every parameter: the cost with the parameter fixed and the other parameters optimized, the threshold of the confidence level and the confidence interval, fit PK_20261020_035132__b4a12. One profile per parameter. A profile which crosses the dashed threshold on both sides gives a finite confidence interval; one which stays below it up to a bound is practically non-identifiable; a flat profile is structurally non-identifiable, the parameter is compensated by the others.

2.5 Provenance

Table 5: Provenance of the fit PK_20261020_035132__b4a12: the versions it ran with and the revision of its models and data; the SHA-256 checksums of the models and data are listed in Table 1.

Item	Value
Fit	`PK_20261020_035132__b4a12`
Created	2026-10-20 01:51:35 UTC
Seed	1234
Python	3.14.4
sbmlsim	0.10.0
libRoadRunner	2.10.0
SciPy	1.18.1
NumPy	2.5.3
Repository	`/home/mkoenig/git/pkdb_models-chlorzoxazone`
Commit	`9223ef2853a750b781a98a244c84bf997b40fdec`
Uncommitted changes	no

Bibliography

[1] M. König, "sbmlsim: SBML simulation made easy." Zenodo, 2026. doi: 10.5281/zenodo.3597770. Version 0.10.0.

[2] R. Storn and K. Price, "Differential evolution - a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997, doi: 10.1023/A:1008202821328.

[3] P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2. Version 1.18.1.

[4] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979, doi: 10.2307/1268522.

[5] C. Welsh, J. Xu, L. Smith, M. König, K. Choi, and H. M. Sauro, "libRoadRunner 2.0: a high performance SBML simulation and analysis library," *Bioinformatics*, vol. 39, no. 1, p. btac770, 2023, doi: 10.1093/bioinformatics/btac770. Version 2.10.0.

[6] A. C. Hindmarsh *et al.*, "SUNDIALS: suite of nonlinear and differential/algebraic equation solvers," *ACM Transactions on Mathematical Software*, vol. 31, no. 3, pp. 363–396, 2005, doi: 10.1145/1089014.1089020.

[7] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, doi: 10.1109/TAC.1974.1100705.

[8] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, doi: 10.1214/aos/1176344136.

[9] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/S0140-6736(86)90837-8.

[10] A. Raue *et al.*, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi: 10.1093/bioinformatics/btp358.

[11] C. Kreutz, A. Raue, D. Kaschek, and J. Timmer, "Profile likelihood in systems biology," *The FEBS Journal*, vol. 280, no. 11, pp. 2564–2571, 2013, doi: 10.1111/febs.12276.

[12] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021, doi: 10.1016/j.coisb.2021.03.005.

[13] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2. Version 2.5.3.